

NAME: _____

PERIOD: _____

One Interval, Two Cutoffs

AP Statistics · Unit 4 · Topic 4.3

[STAR] THE PROBLEM

Suppose a library wants the mean setup time for all 800 kiosk transactions in a month below 45 seconds and hopes to advertise a mean below 42 seconds. An SRS of 25 has the shown summary. A 95% one-sample t -interval is (39.923, 44.877) seconds.

Mina: “I would compare each cutoff with the full interval and keep the conclusion about the population mean.”

Gabe: “Because the interval contains times below both cutoffs, I think it supports both claims; I also read 95% as the share of transaction times inside it.”

Kiosk sample

Summary	Value
Population	800
n	25
Mean (s)	42.4
SD (s)	6.0
Shape	Symmetric; no outliers

FIRST TAKE — one sentence before you work the parts

Does the range 39.923 to 44.877 make both time claims believable? Give one reason.

[BOOK] LET THE ENTIRE INTERVAL GOVERN THE CLAIM (keep on the page)

A $C\%$ confidence interval estimates a population mean, not the spread of individual observations. We are $C\%$ confident that the interval captures the population mean represented by the random sample; the fixed interval either contains that mean or it does not. A value inside the interval remains plausible. To support $\mu < c$, the entire interval must lie below c ; an endpoint merely falling below c is not enough. With other features held fixed, a larger sample tends to narrow a mean interval, while a higher confidence level widens it.

Work (a)–(c).

DOK 1 (a) Locate 42 and 45 relative to the interval.

DOK 2 (b) Interpret the supplied 95% interval in one sentence about the mean setup time for all 800 transactions.

DOK 3 (c) In 3–4 sentences, decide whose approach is justified. Use the interval to judge each cutoff, then explain what Gabe gets wrong about individual transaction times.

Frame: Relative to the interval, 42 is _____ and 45 is _____; together, those locations support _____.

Frame: The interval estimates _____ rather than _____, so the advertisement should _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.